
paper_id: PAPER_1004
 title: "QGP Vacuum Density with SCm $S_{26}^{(3)}$ Phonon Coupling"
 session: 219
 date: 2026-04-13
 author: "Daniel T. Murphy"
 status: production
 cvw: "v2.0.0"
 tags: [QGP, vacuum, density, SCm, phonon, deconfinement, quark]
 crosslinks: [PAPER_969, PAPER_999, PAPER_1005]
 calibration: {T_c: 1.5e12, rho_QGP_2e12: 1.16e-12, S26_3rd: 0.095}
 sm_anchor: "CVW v2.0.0 — G6 SM Anchor Gate compliant"

PAPER_1004: QGP Vacuum Density with SCm Phonon Coupling

Abstract

We compute the QGP vacuum density $\rho_{\text{QGP}}(T)$ through SCm phonon coupling using $S_{26}^{(3)}$. Below $T_c = 1.5 \times 10^{12}$ K, confinement suppresses the QGP state; above T_c , the density follows exponential activation with Gaussian phonon response.

1. Core Equation

$$\rho_{\text{QGP}}(T) = \rho_{\text{SCm}} \cdot S_{26}^{(3)} \cdot \exp(-(T_c - T)/T) \cdot \Phi(T)$$

$$\text{where } \Phi(T) = S_{26}^{(3)} \cdot \exp(-(T - T_c)^2 / (2(0.1T_c)^2)).$$

2. Results

Temperature	ρ_{QGP}
$T = 10^6$ K	0 (confined)
$T = 2 \times 10^{12}$ K	1.16×10^{-12} kg/m ³
$T = 5 \times 10^{12}$ K	$\sim 10^{-15}$ kg/m ³

3. Implementation

File: scm_qgp_dynamics.py, class QGPVacuumDensityCalc. CP4 class #588.